

Lab 1: Setup and Instruction Set

- **Link**

- Download and install MPLAB X IDE : <https://youtu.be/2Y8gKUcDrnc>
- Introduction to Instruction Set : <https://youtu.be/LsTaLYXPXCQ>
- HackMD : <https://hackmd.io/@mprocessor2026/SyWBGkYrGx>
- HackMD : <https://hackmd.io/@mprocessor2026/ryprf1FSMg>

- **Basic (50%)**

- **Description**

Store four unsigned 8-bit values in memory: x1 at address [0x000], x2 at address [0x001], y1 at address [0x002], and y2 at address [0x003]. First, compute $A1 = x1 + x2$ and store the result at address [0x010]. Next, compute $A2 = y1 + y2$ and store the result at address [0x011]. After completing both additions, compare A1 and A2:

- If $A1 > A2$, store 0x11 at [0x020].
- If $A1 = A2$, store 0x22 at [0x020].
- If $A1 < A2$, store 0x33 at [0x020].

You may assume that **neither** addition operation will cause an overflow, i.e., $x1 + x2 \leq 0xFF$ and $y1 + y2 \leq 0xFF$.

- **Example**

Address	Testcase 1	Testcase 2
[0x000]	0x12	0x0C
[0x001]	0x08	0x14
[0x002]	0x0A	0x18
[0x003]	0x05	0x08
[0x010]	0x1A	0x20
[0x011]	0x0F	0x20
[0x020]	0x11	0x22

- **Standard of grading**

1. Your program must use at least one of the following PIC18F comparison instructions when comparing A1 and A2: **CPFSEQ**, **CPFSGT**, or **CPFSLT**. For addition and subtraction operations in your design, you may also use **ADDWF**, **SUBWF**, or other similar

instructions as needed.

- The results must be stored in the correct memory locations.

- **Advance (30%)**

- **Description**

An 8-bit unsigned value is stored at data memory address [0x000]. Design a loop that reverses the order of all eight bits in this value. Specifically, if the original bit sequence is b7 b6 b5 b4 b3 b2 b1 b0 the reversed sequence must be: b0 b1 b2 b3 b4 b5 b6 b7. Store the reversed 8-bit result at data memory address [0x010]. After obtaining the reversed value, compare the reversed result with the original value stored at address [0x000]:

- If they are equal, the original number is a bit palindrome. Store 0xFF at address [0x011].
- If they are not equal, store 0x00 at address [0x011].

The original value stored at address [0x000] must remain **unchanged** throughout program execution.

- **Example**

Address	Testcase 1	Testcase 2
[0x000]	b'11010010'	b'10011001'
[0x010]	b'01001011'	b'10011001'
[0x011]	0x00	0xFF

- **Standard of grading**

- Your program must use a **loop** to process and reverse all eight bits of the input value. Brute-force approaches, including lookup tables, case-by-case enumeration, or separately handling all possible input values, are not allowed.
- The reversed value and the palindrome-detection result must be correctly stored at addresses [0x010] and [0x011], respectively. The original value at address [0x000] must not be modified.

- **Hard (20%)**

- **Description**

Store an 8-bit unsigned number at memory location [0x000], and design a loop to find the longest sequence of consecutive 1 bits in the number. For example, consider the following input: b'10111011'. The consecutive sequences of 1 bits are: 1, 111, 11. The longest sequence contains **three** consecutive 1 bits. Therefore, the result is: 0x03. Store the length of the longest sequence of consecutive 1 bits at data memory address [0x010].

The original value stored at address [0x000] must remain **unchanged**

throughout program execution.

■ **Example**

Address	Testcase 1	Testcase 2
[0x000]	b'10111011'	b'01111100'
[0x010]	0x03	0x05

■ **Standard of grading**

1. The final result must be correctly stored at address [0x010], and the original value at address [0x000] must not be modified.
 2. Your program must use a **loop** to examine all eight bits of the input value. Brute-force approaches, including lookup tables, case-by-case enumeration, or separately handling all possible input values, are not allowed.
 3. The implementation must use one or more appropriate PIC18F rotation or shift instructions, such as **RRCF**, **RRNCF**, **RLCF**, or **RLNCF**.
- Note: If you have any questions, please send an email to the TA mailbox (mprocessor2026@gmail.com). Emails sent to individual TAs will not be answered.